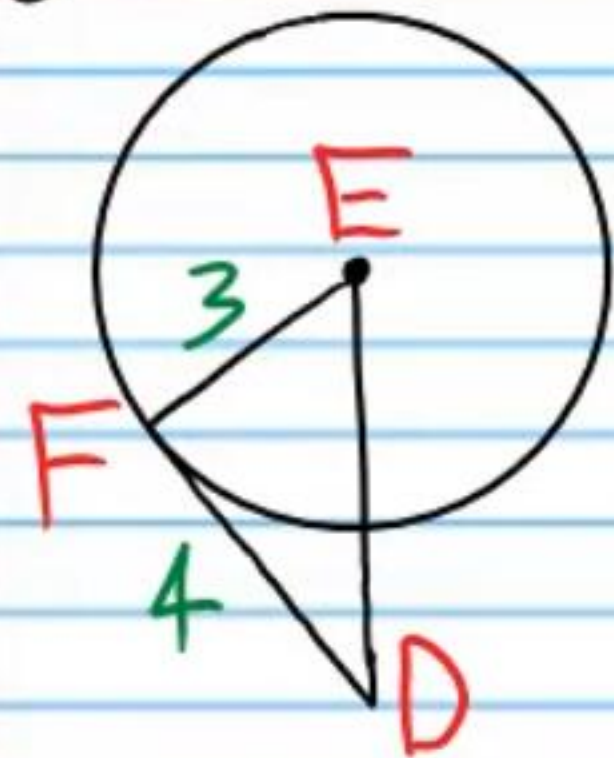


Circles (Part I): Tangents and Arcs (Homework)

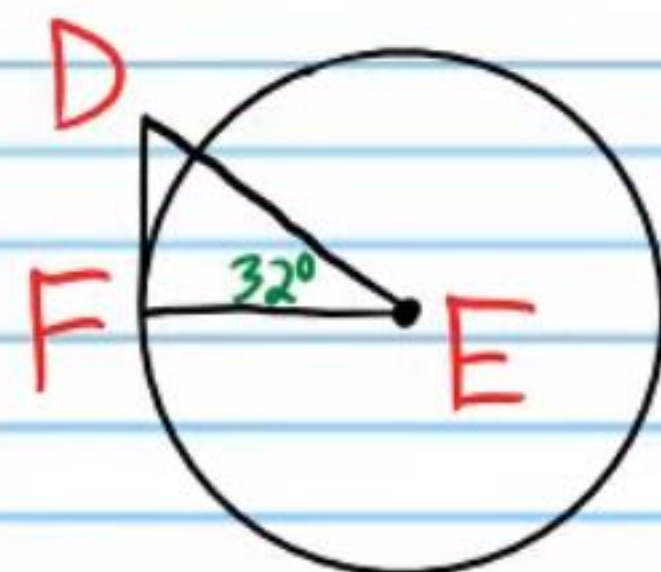
If the line containing $\overline{FD}$ is tangent to $\odot E$ at F , find the following values.

① ED



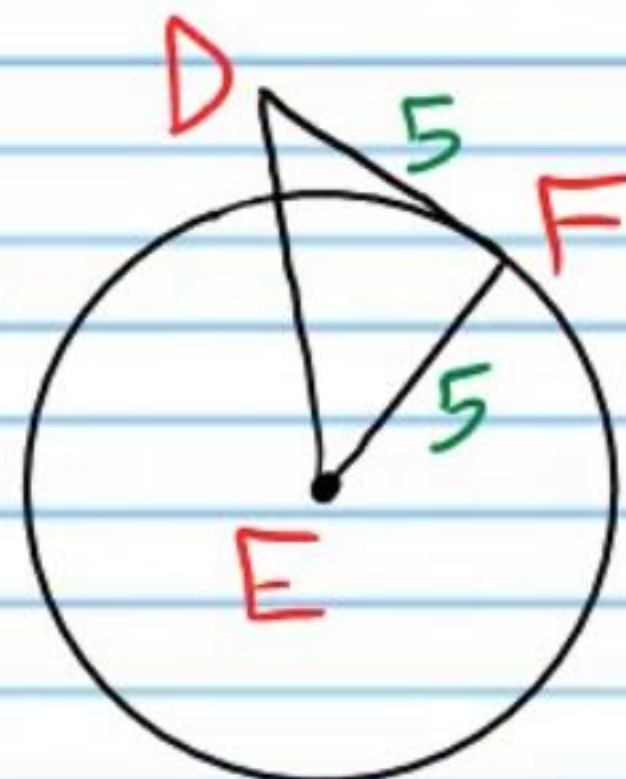
5

② $m\angle D$



58°

③ DE & $m\angle D$

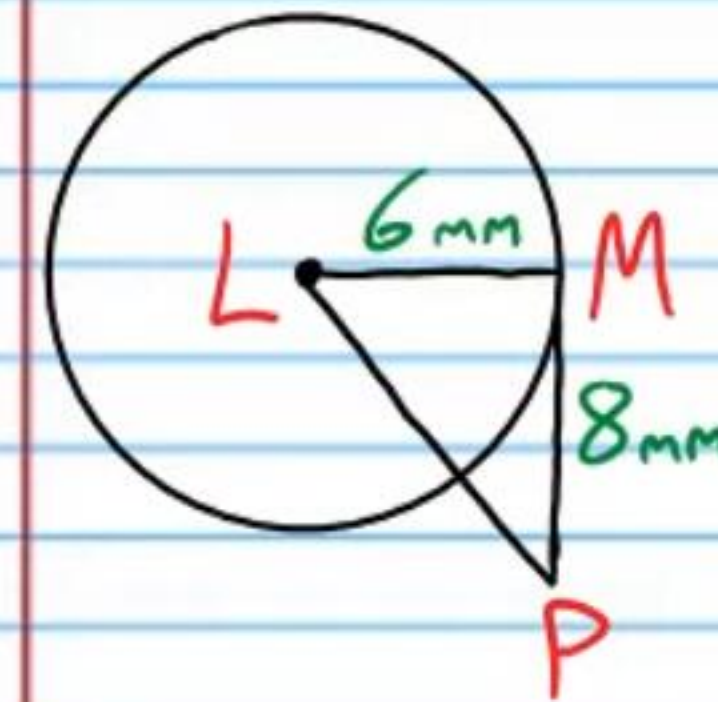


$$DE = 5\sqrt{2}$$

$$m\angle D = 45^\circ$$

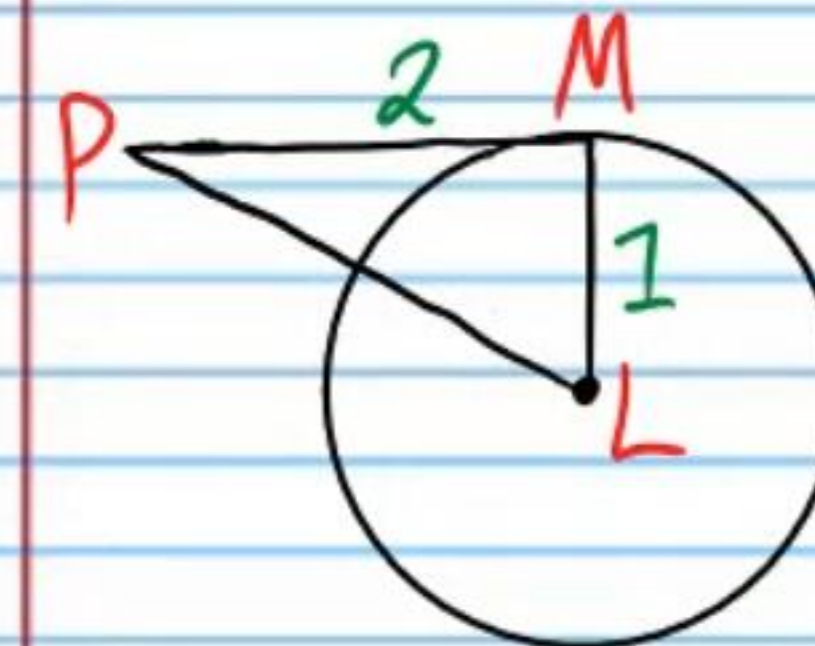
If $\overline{LM}$ is a radius, is the line containing $\overline{MP}$ tangent to $\odot L$? Justify your answer.

④ $PL = 12 \text{ mm}$



No

⑤ $PL = \sqrt{5}$



Yes

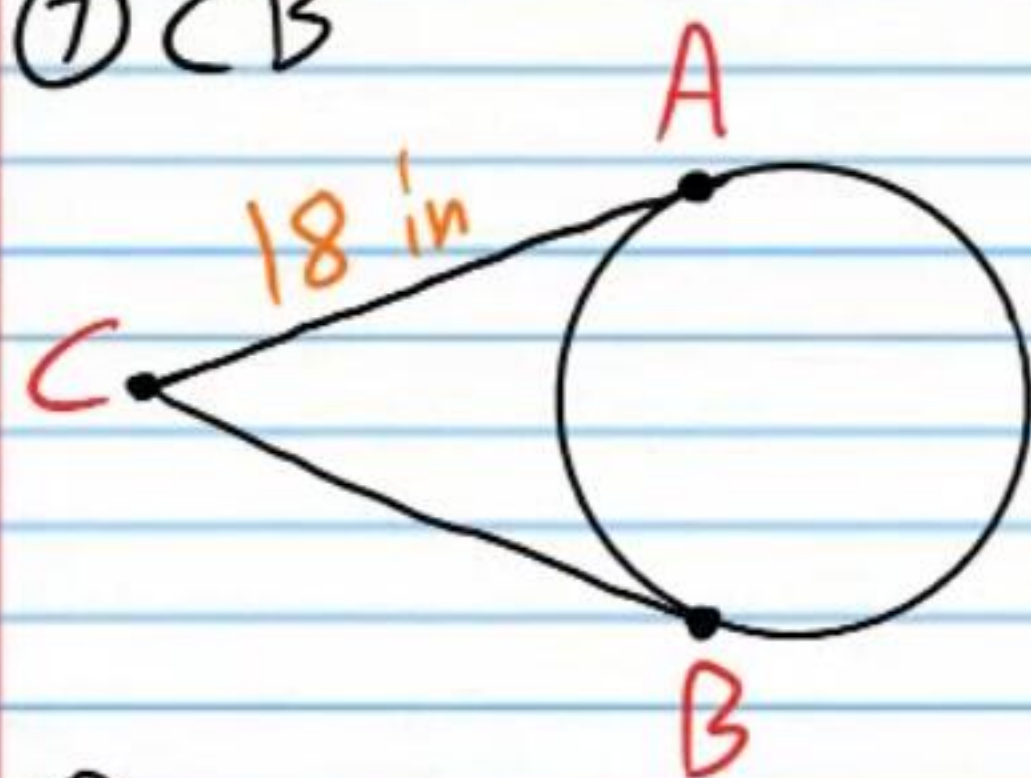
⑥ $PL = 3$



No

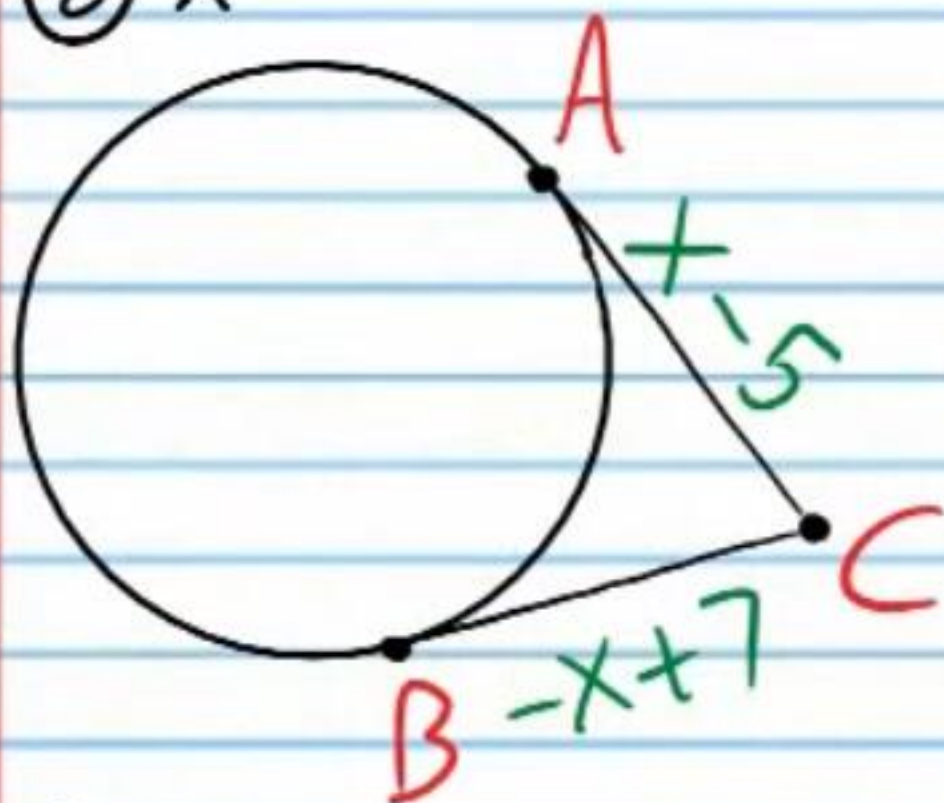
If A and B are points of tangency, find the following values.

⑦ CB



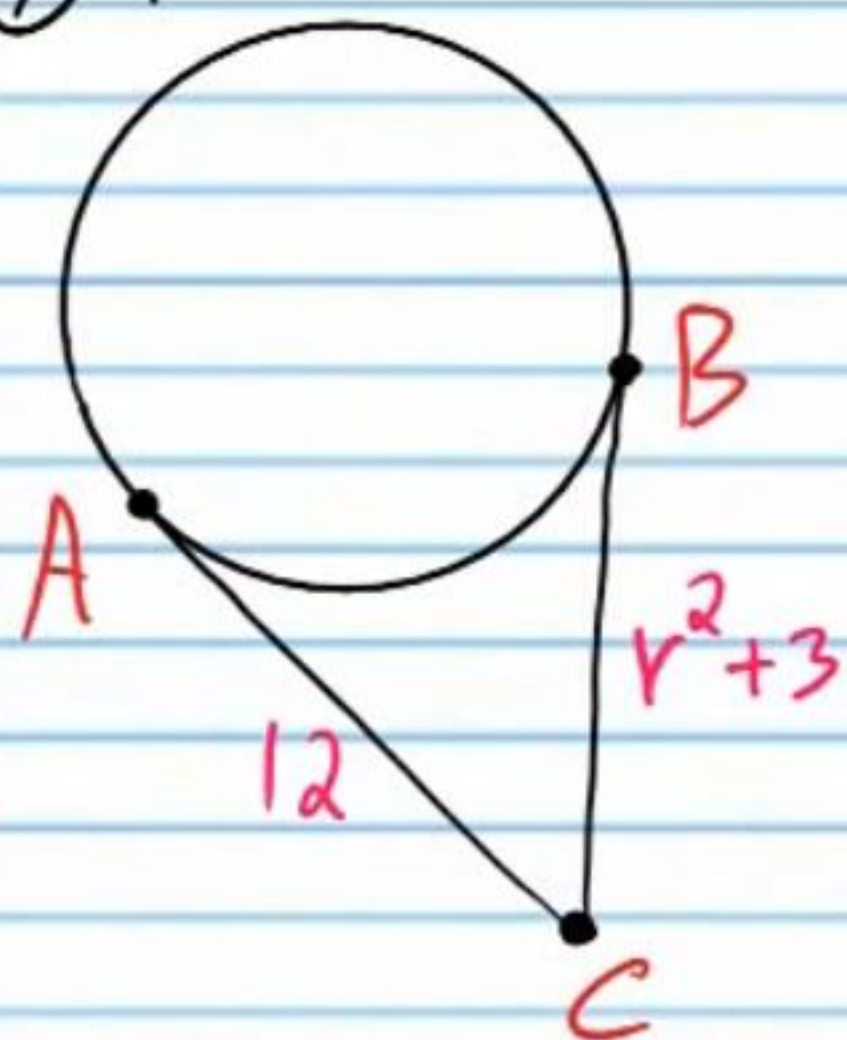
18 in

⑧ x



6

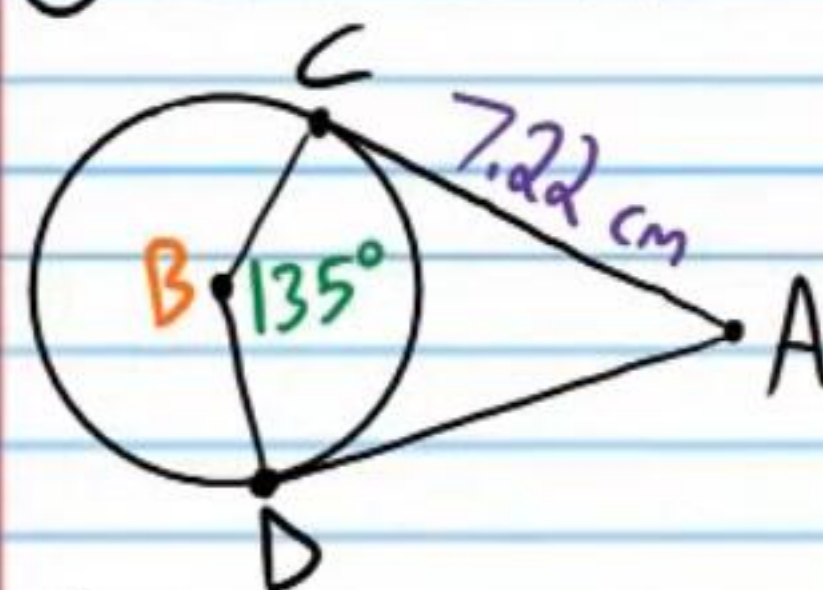
⑨ r



±3

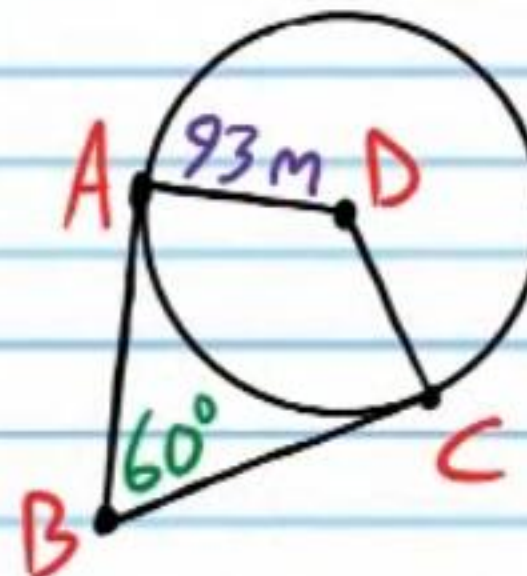
If the segments below are radii and tangents, find the following values.

⑩ $m\angle A$ & AD.



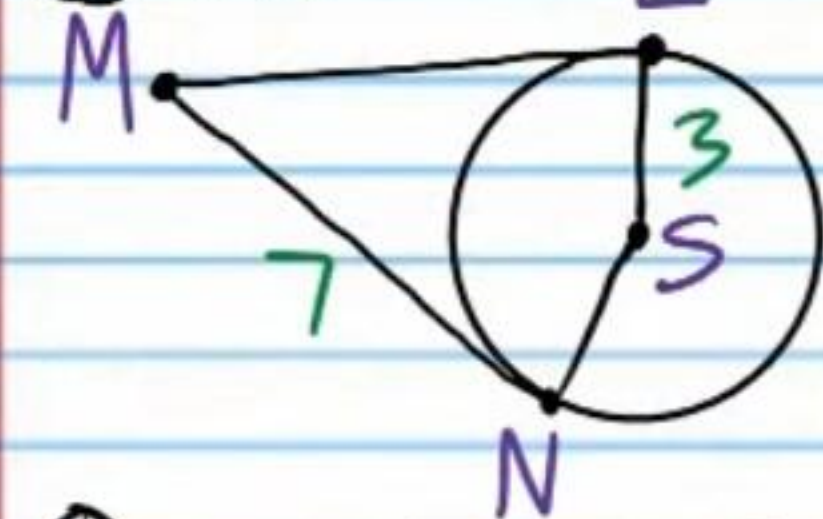
$m\angle A = 45^\circ$
 $AD = 7.22 \text{ cm}$

⑪ DC & $m\angle D$.



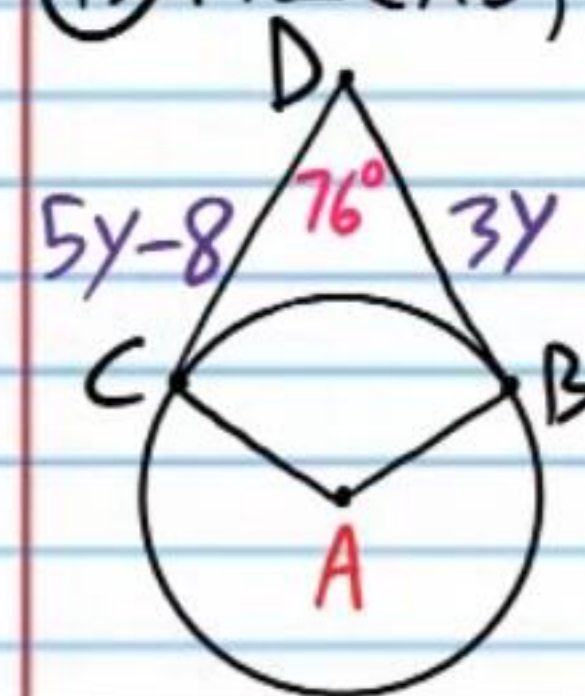
$DC = 93 \text{ m}$
 $m\angle D = 120^\circ$

⑫ MS



$\sqrt{58}$

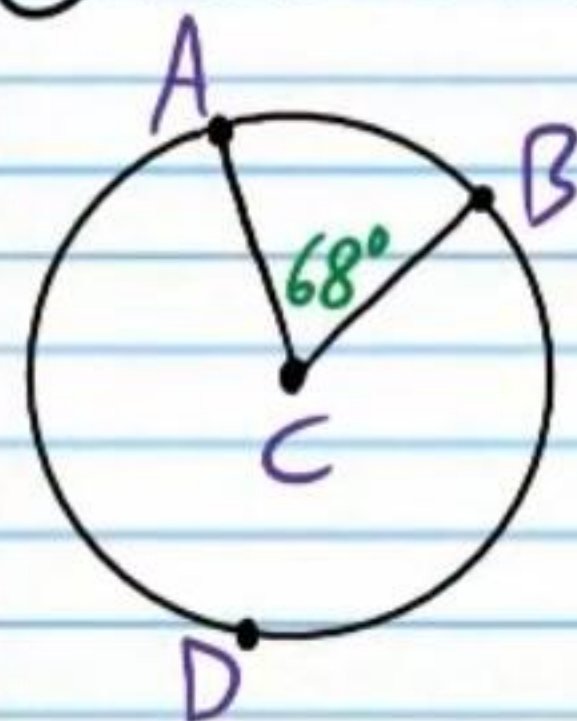
⑬ $m\angle CAD$, $m\angle A$, & y.



$m\angle CAD = 52^\circ$
 $m\angle A = 104^\circ$
 $y = 4$

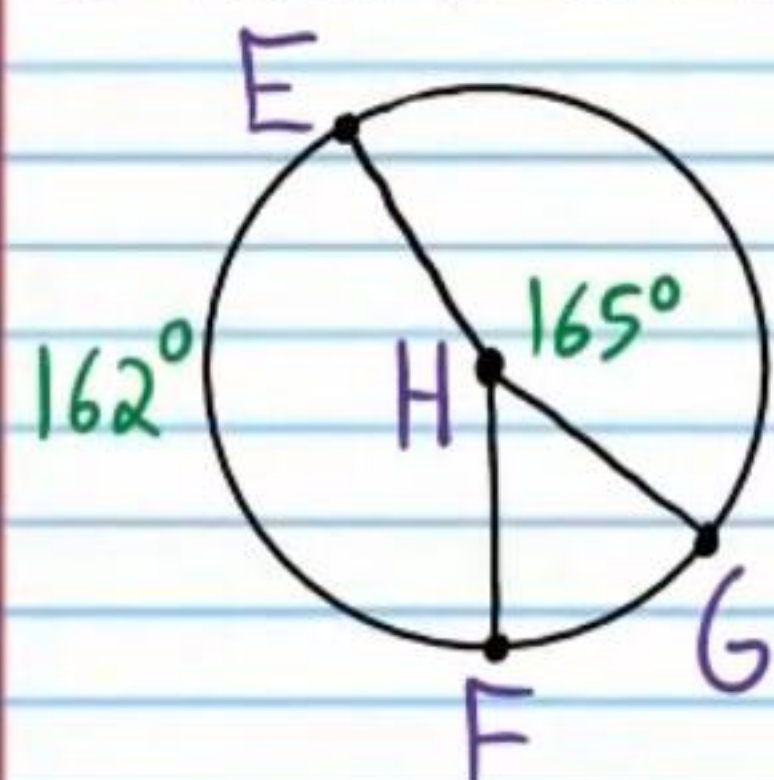
If the points inside the circles below are the centers of the circles, find the following values.

⑭ $m\widehat{AB}$ & $m\widehat{BDA}$



$$\begin{aligned} m\widehat{AB} &= 68^\circ \\ m\widehat{BDA} &= 292^\circ \end{aligned}$$

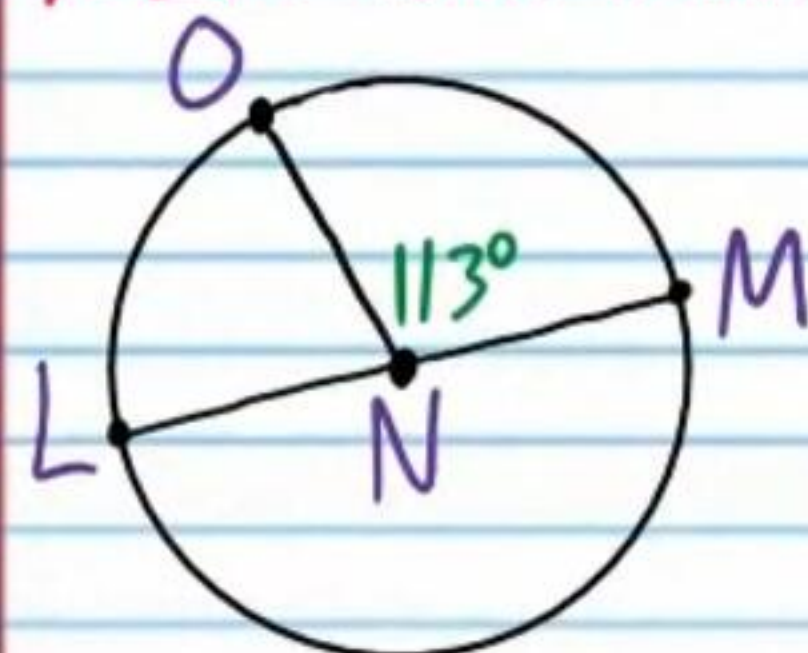
⑮ $m\widehat{EFG}$ & $m\angle FHG$



$$\begin{aligned} m\widehat{EFG} &= 195^\circ \\ m\angle FHG &= 33^\circ \end{aligned}$$

⑯ $m\widehat{LO}$ & $m\angle LNO$

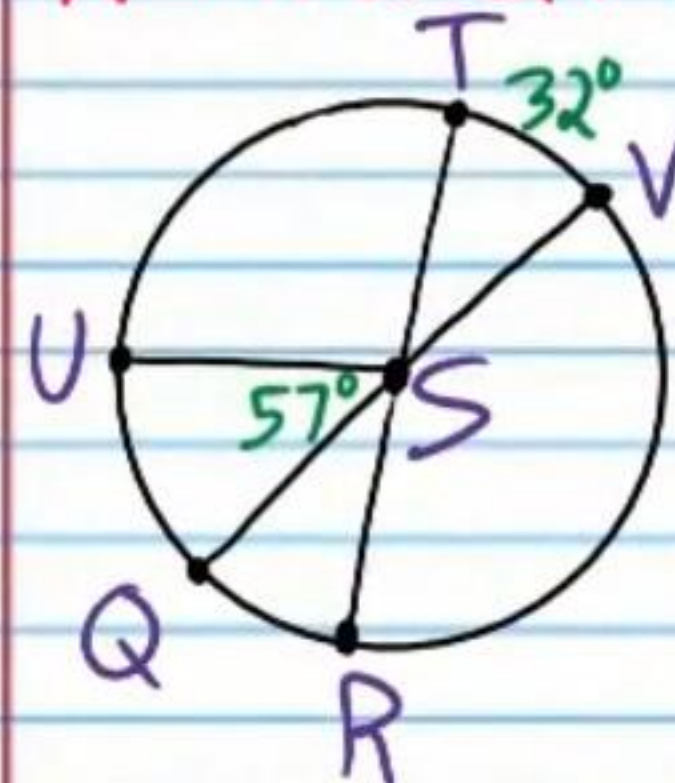
$\overline{LM}$ is a diameter.



$$\begin{aligned} m\widehat{LO} &= 67^\circ \\ m\angle LNO &= 67^\circ \end{aligned}$$

⑰ $m\angle UST$ & $m\widehat{RV}$

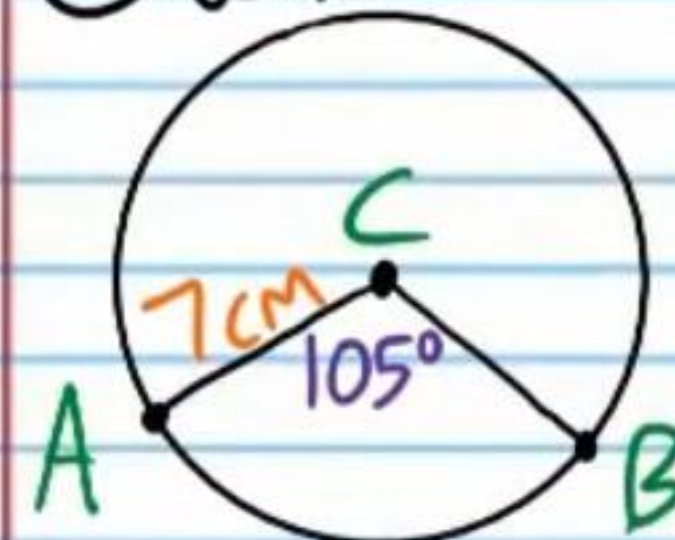
$\star$ $\overline{TR}$ & $\overline{QV}$ are diameters.



$$\begin{aligned} m\widehat{RV} &= 148^\circ \\ m\angle UST &= 91^\circ \end{aligned}$$

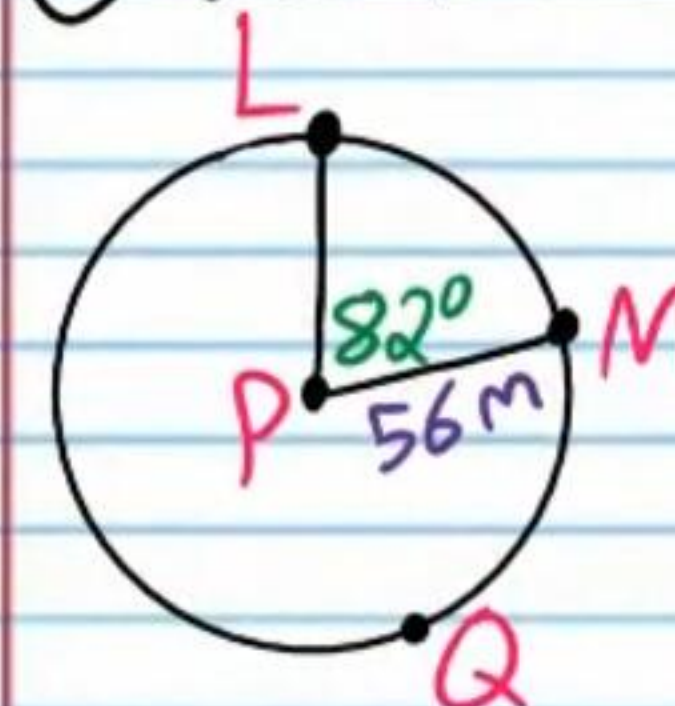
Find the lengths of the following arcs. Round to the nearest hundredth. $\star$ You must use the π button on your calculator. The points inside the circles are the centers of the circles.

⑱ $\mathcal{L}\widehat{AB}$



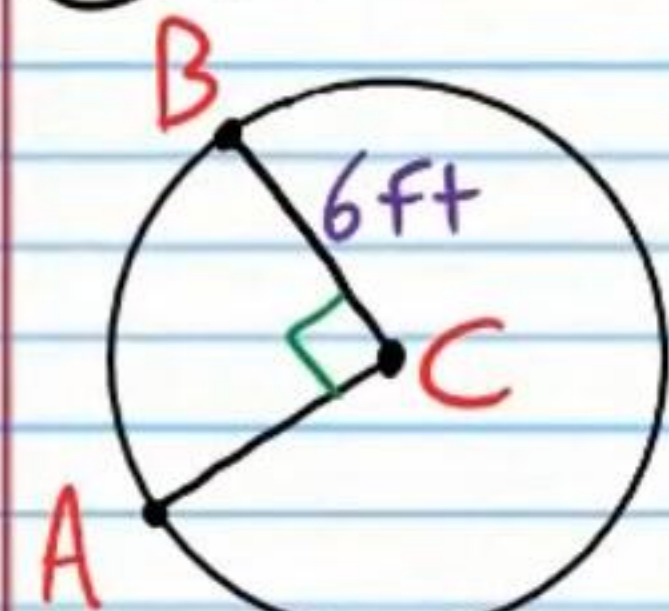
$$12.83 \text{ cm}$$

⑲ $\mathcal{L}\widehat{LQM}$



$$271.71 \text{ m}$$

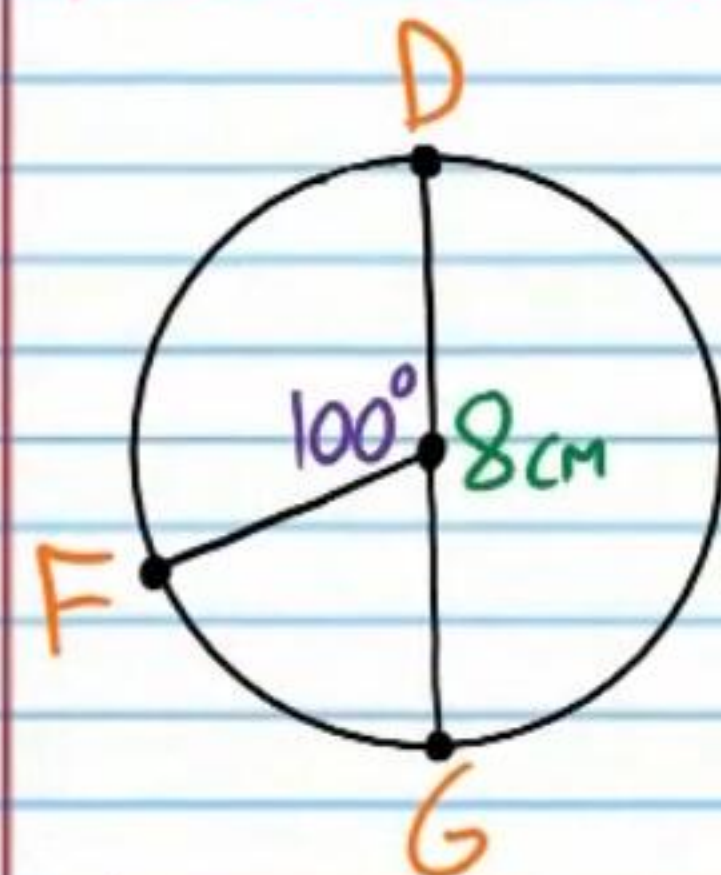
20) $\angle \widehat{AB}$



9.42 ft

21) $\angle FGD$

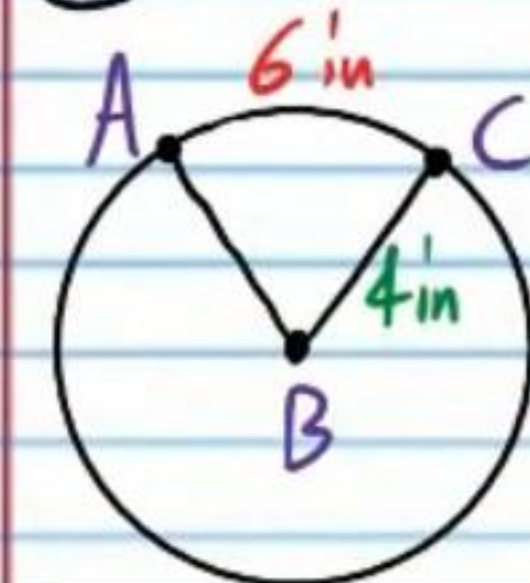
$\star \overline{DG}$ is a diameter.



18.15 cm

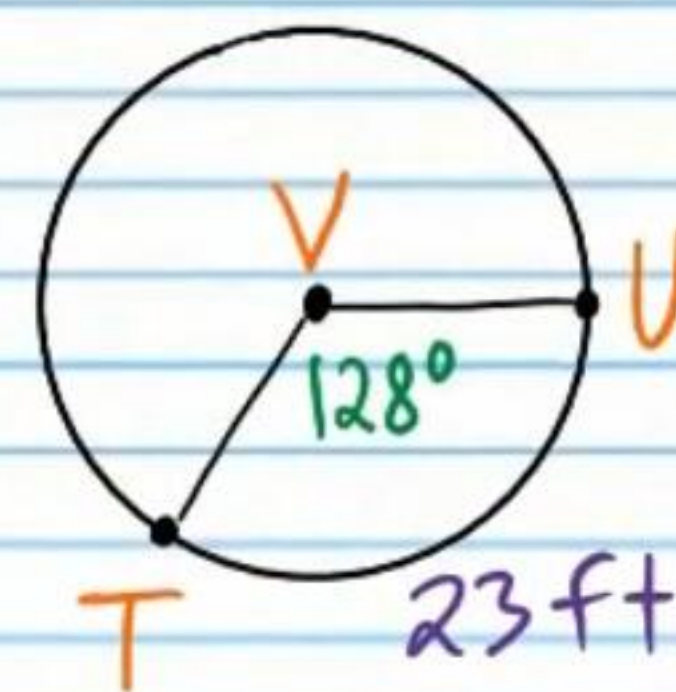
Find the following values. Round to the nearest hundredth.
 $\star$ You must use the π button on your calculator.
 The points inside the circles are the centers of the circles.

22) $m\angle ABC$



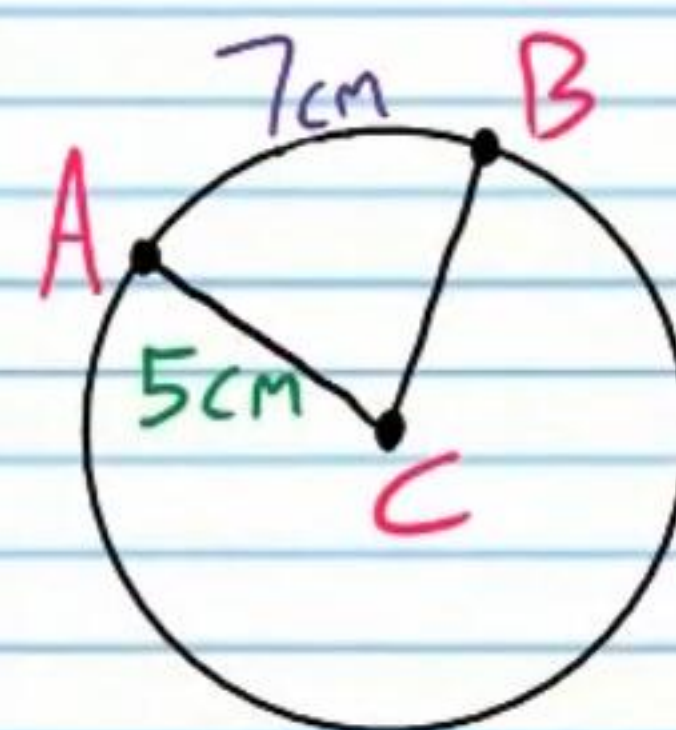
85.94°

23) $\widehat{VU}$



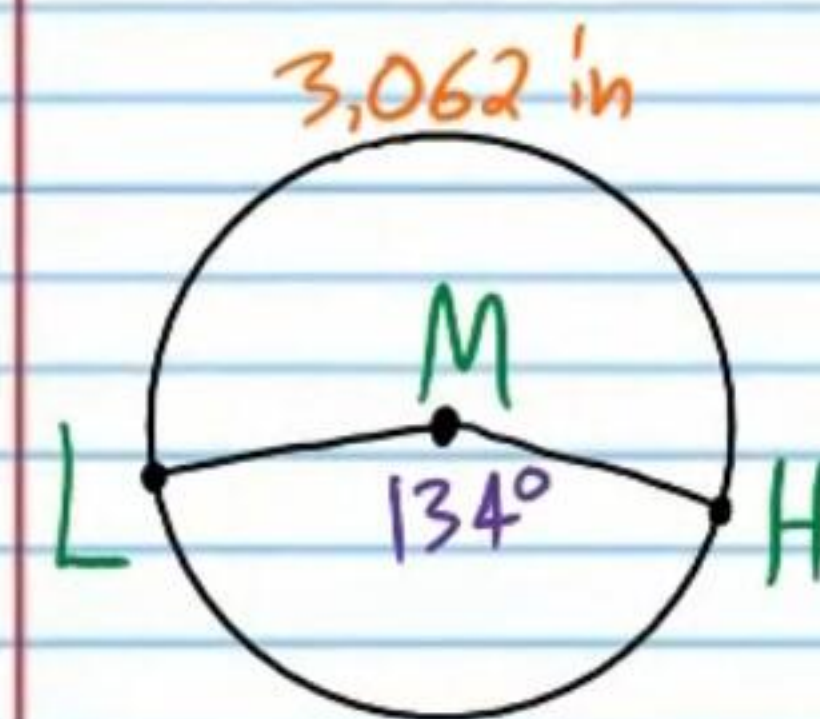
10.3 ft

24) $m\widehat{AB}$



80.21°

25) Circumference



4,877.52 in